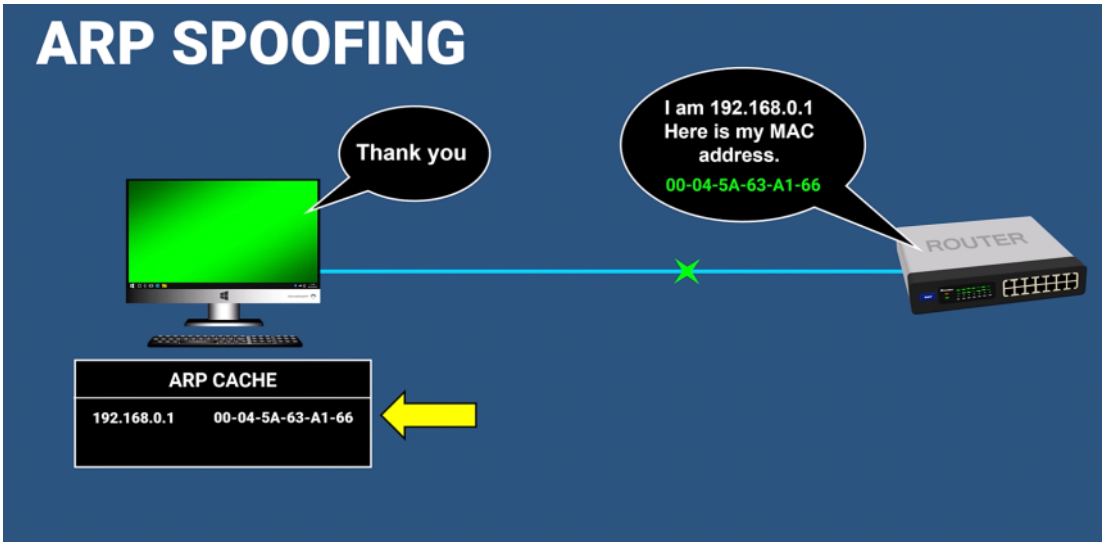
ARP Spoofing

Sunday, March 1, 2026 10:13 PM

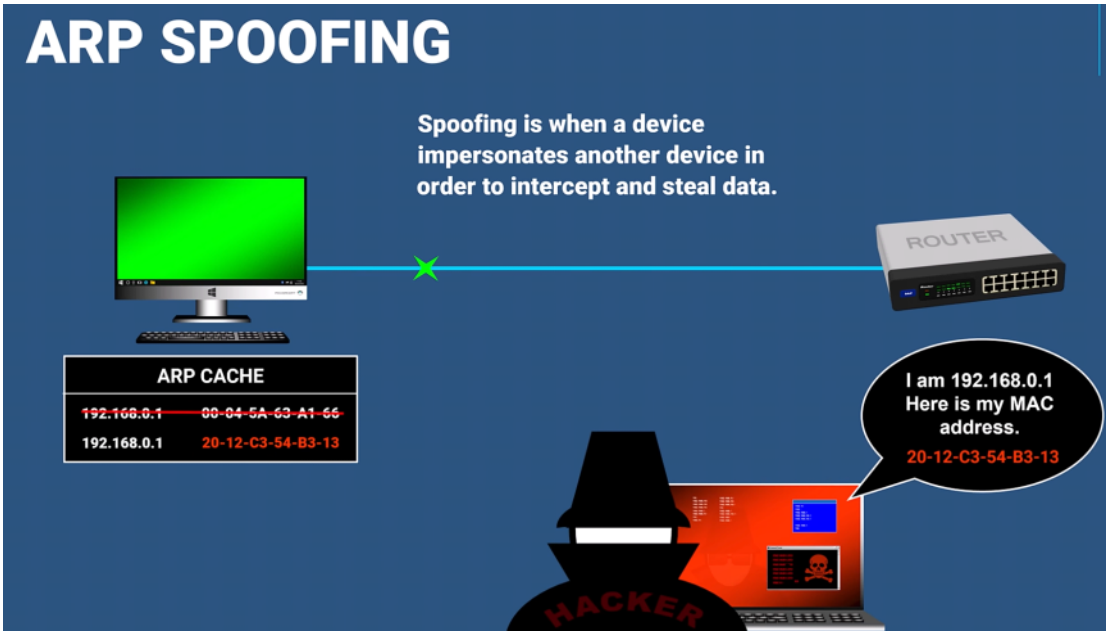
ARP (Address Resolution Protocol):

A network protocol used to map or resolve an IP address to a MAC address.
The MAC address is a unique physical identifier assigned to every network interface card (NIC).
Devices use MAC addresses to communicate at the data link layer within a local network.
Normal ARP Process:



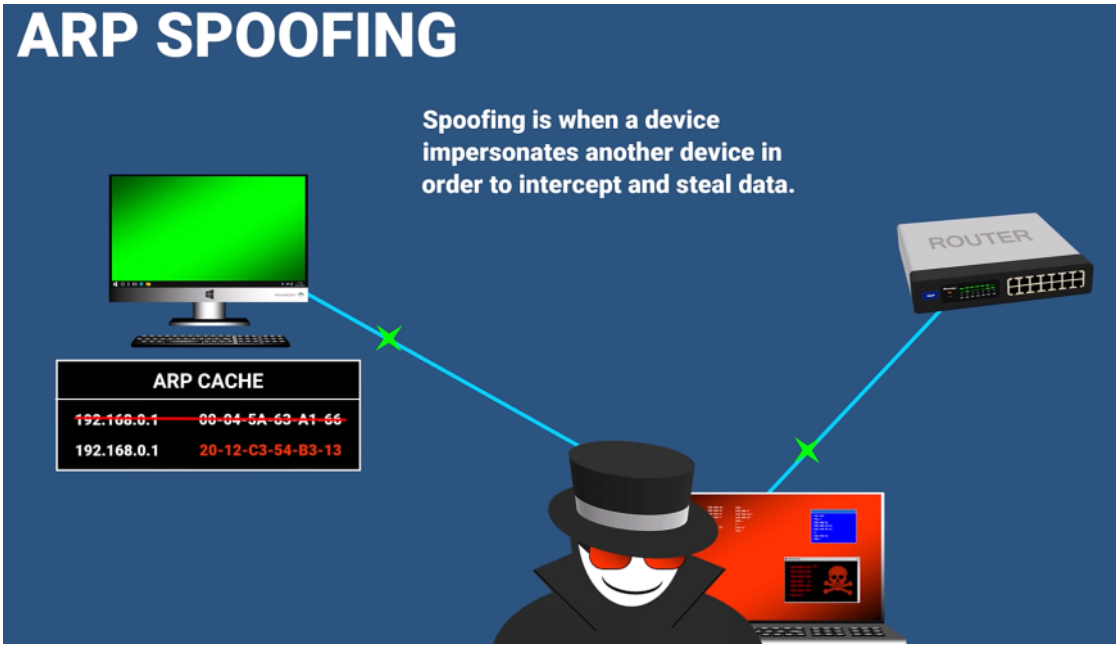
When a device (e.g., a computer) wants to send data to another device (e.g., a router), it needs the target device's MAC address.
The computer broadcasts a request on the network asking, "Who has this IP address?"
The device owning that IP (the router) replies with its MAC address.
The computer stores this IP-to-MAC mapping in its ARP cache for future communication.
ARP Spoofing Explained
Definition:

ARP spoofing is an attack where a malicious device impersonates another device on the local network by sending fake ARP responses.
This fake response misleads the victim device into associating the attacker's MAC address with the IP address of a legitimate device (usually the router).
Attack Process:



The attacker joins the same network as the victim.
The attacker's device sends false ARP replies claiming to have the router's IP address but with the attacker's MAC address.
The victim updates its ARP cache with the attacker's MAC address, believing it to be the router's.

Impact:



All traffic intended for the router is first sent to the attacker's device.
The attacker can intercept, modify, or steal the data before forwarding it to the router.
This enables a Man-in-the-Middle (MitM) attack, also called an on-path attack, where the attacker secretly relays and possibly alters communication between two parties.